

PRODUCT MANAGER TEST ASSIGNMENT · NOMAD VENTURE STUDIO

Jobescape.

The assignment, task by task.

4 parts · 11 tasks · 9,956 paying subscribers

ALIASKAR BEKISHEV · JULY 2026

Eleven tasks.

Brief's own order · summary p.31 · references p.33 · appendix p.34–46

A1	Audience segments — what I split by	07–08
A2	Product vs need — the gap, the risk	09–10
B1	Competitors — direct vs indirect	12
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B3	What to take, and the order	14
C1	Release analytics — high / low / didn't take	16–21
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C4	Prototype	26
D1	Net LTV per plan, blended	28
D2	A/B model, break-even, recommendation	29–30

Computed, not quoted.

5 SQL queries

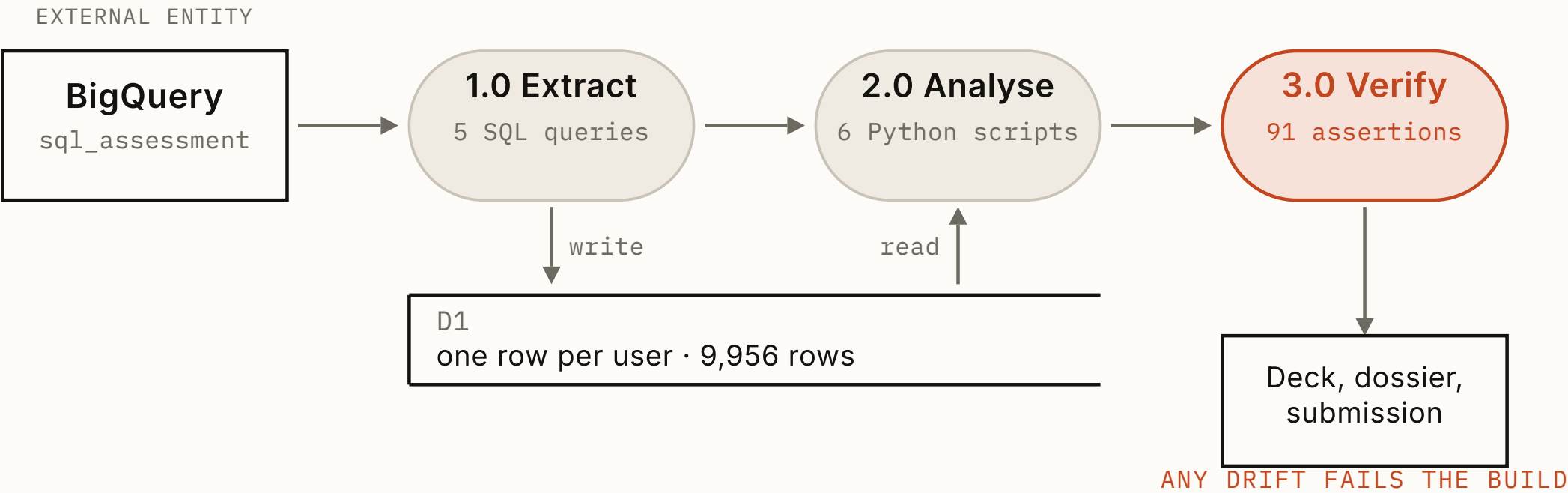
one row per user

6 Python scripts

pure standard library

91 assertions

re-derived from the raw CSV¹



IF YOU REMEMBER ONE THING

50.2%

OF PAYING SUBSCRIBERS NEVER COMPLETE A
SINGLE LESSON – ANYWHERE²

Selling	excellent
Activation	never built
Economics	depend on the tail activation would protect

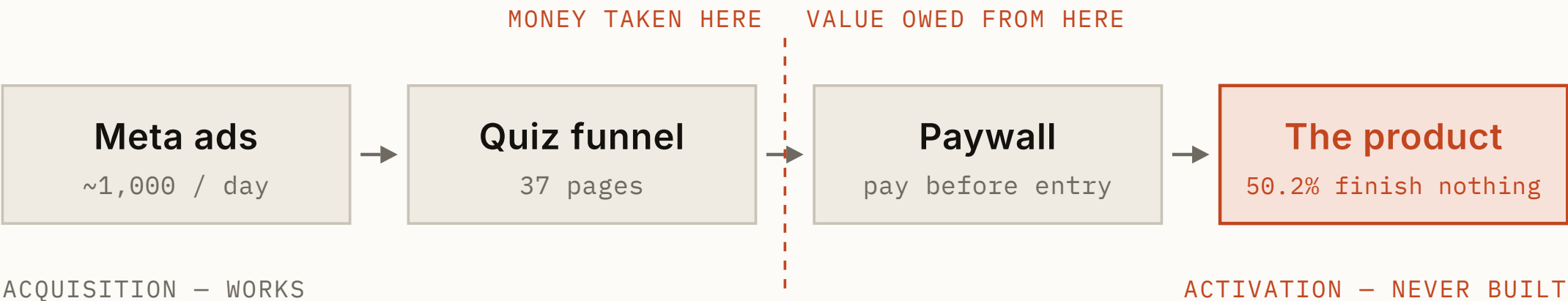
Money first. Value later.

Paywall-first

every user in the data is a payer³

Consequence

failed activation = refunds, not lost signups



PART A

Audience & product

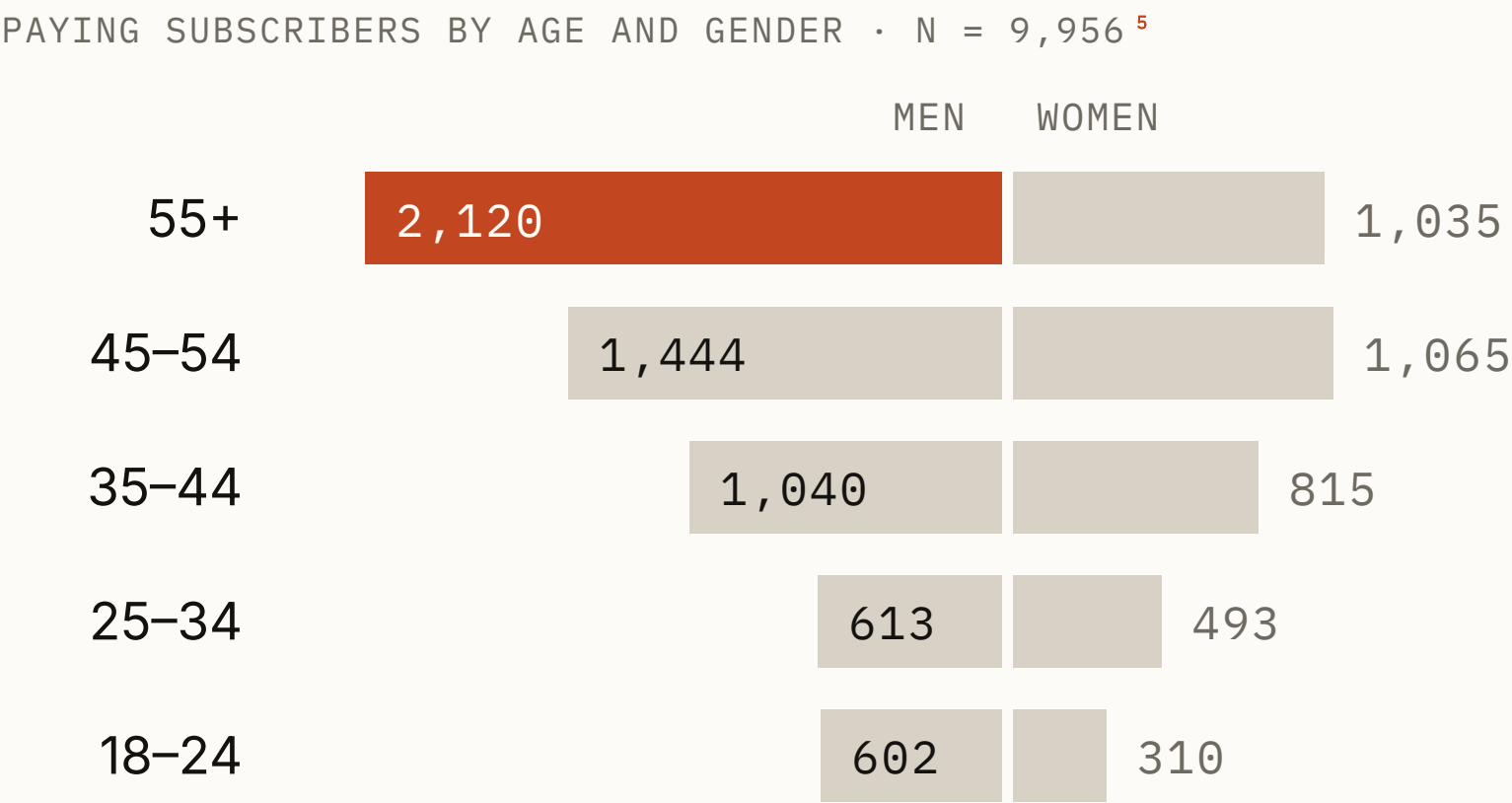
Clustered, not asserted.

SEGMENT	WHO THEY ARE	SHARE	STRONGEST SIGNAL
The Striver	senses a gap, closing it deliberately	28.4%	"could be behind" 1.9×
The Latecomer	oldest, hasn't started, watching others pull ahead	22.0%	"falling behind" 2.2×
The Adept	already gets real value from AI, wants more	20.9%	"AI already helps me" 2.5×
The Optimiser	mid-career, uses Claude on work, wants speed	12.3%	aged 35–44 2.6×
The Founder	learning AI to build something of their own	9.0%	"start my own business" 3.1×
The Switcher	between jobs, starting from zero	7.4%	"exploring options" 4.4×

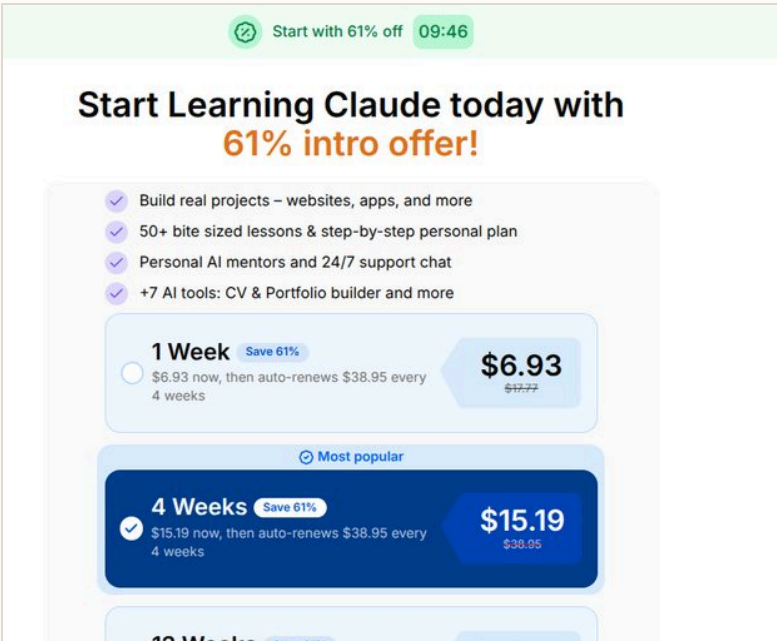
k-modes on 38,071 quiz respondents × 10 answers. Signal = lift vs population. **No elbow** — this is a continuum, so six is a resolution I chose, not a count the data forced.⁴

A decade older than the internal read.

- Gender read right — 59.7% men
- Age read ~60% are 45+
- Biggest cell men 55+
- Churn 31.1% at 18–24 → 9.5% at 45–54



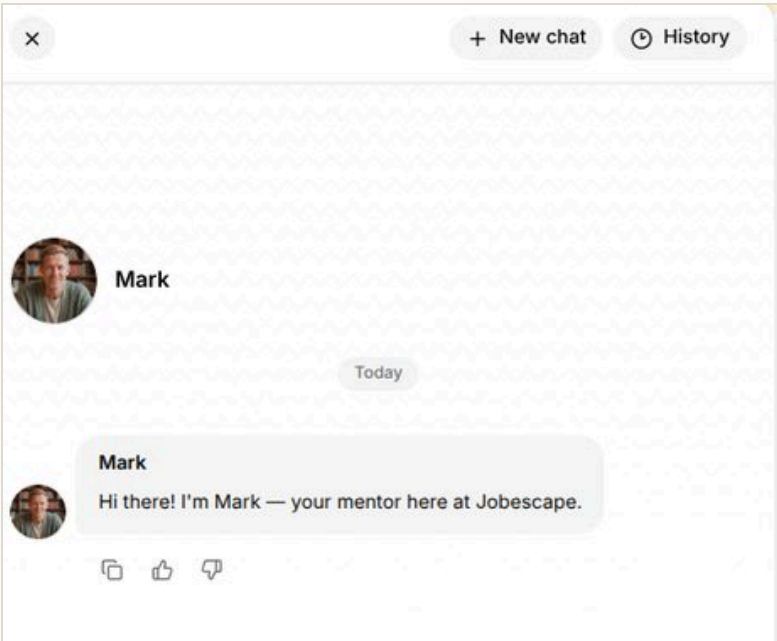
Paid. Walked it. Checked.



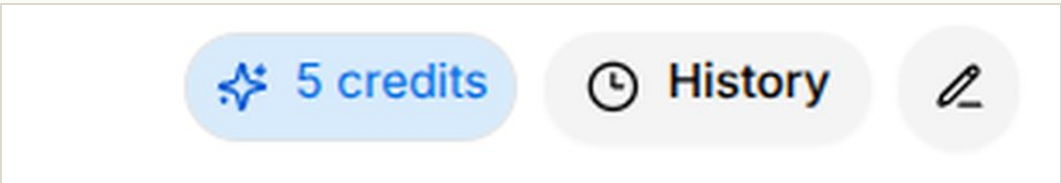
PROMISED AT THE PAYWALL

- Sold** “personal AI mentors · 24/7 support chat”
- Got** one bot, stock photo, named Mark
- Got** chat metered to 5 credits

My own read corrected: cancellation is **self-serve** — 1,358 uses. The trap is the refund. ⁶



THE “MENTOR”



THE “24/7 CHAT”, METERED

11.3%

OF UNSUBSCRIBES ARE PAYMENT FAILURES,
CHARGEBACKS OR DISPUTES⁷

Scale	157 of 1,393 · in 14 days
Mechanism	refund gated on finishing in ~31 days
Exposure	the model FTC sued Genesis over, June 2026 ⁸

PART B

Competitors

4 direct twins. The dangerous one is indirect.

Same job use AI at work without feeling behind

Decisive the content itself is free

So win on scaffolding, feedback, accountability⁹

DIRECT

Coursiv
~95% overlap · identical funnel

Iro AI
“Duolingo for AI” · weak commercially

Outskill
same buyer · premium price

Be10x
tripwire CRO · US launch pending

INDIRECT

Section AI
pivoted courses → AI co-pilot

The Rundown
2M newsletter → \$1,008/yr

Mimo / Headway
the paywall playbook

ChatGPT, YouTube
give it away free

Nobody wins on content.

COMPETITOR	OVERLAP	MECHANIC	WELL / BADLY
Coursiv	~95%	Finite branded challenge · honest pricing	Container works · content thin
Iro AI	High	Graded Prompt Lab	Real practice · no distribution
Section AI	High	Role-specific co-pilot, saveable	Admits courses don't retain
Mimo	Adjacent	Trial-timeline paywall + reminder	Best paywall · off-subject
The Rundown	Broad	Newsletter → paid university	Owned audience · not skills
From public product, pricing pages and trials. ¹⁰			

Order follows the money.

#	STEAL	FROM	WHY HERE
1	Daily-cost paywall with anchored discount	Mimo	highest ROI, lowest cost
2	Branded finite challenge as the container	Coursiv	retention, and you half-built it
3	A practice surface — prompt lab, saveable output	Iro · Section	the only durable moat
4	Owned newsletter / in-app daily brief	The Rundown	compounds, so start it early
5	Free-workshop → high-ticket ladder	Outskill	last — needs retention first

De-risk → retain → differentiate → diversify → expand.¹¹

Both are container work. Neither is content.

You do not need better lessons. You need better vessels for the ones you have — and vessels are the cheap half.

-
- Steal 1 · the paywall

day-price framing ("\$.71/day"), the original struck through, one pre-selected plan. Mimo ships it; your paywall sells a 61% discount **with no daily anchor**
-
- Steal 2 · the container

a branded, finite, gated challenge — the thing Part C proves you half-built. Coursiv brands theirs and gates the days
-
- Why these two first

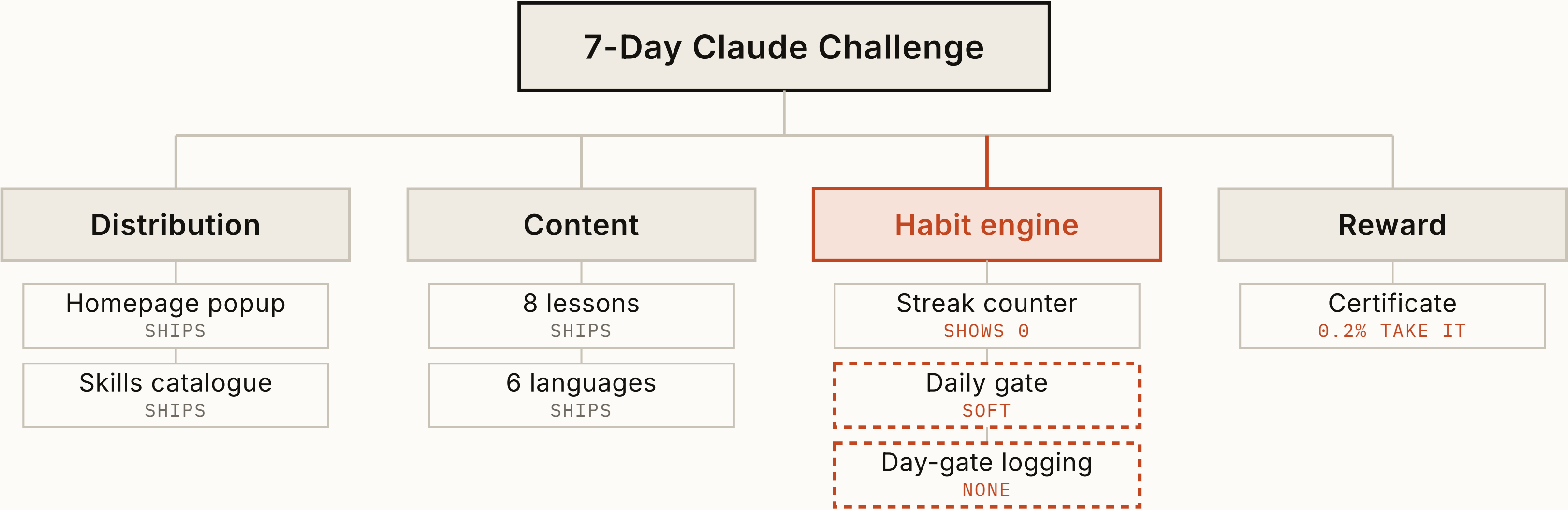
both defend revenue already booked. Everything else on the list chases revenue you do not have yet ¹⁰

PART C

The Challenge release

The gate is a suggestion.

One day = one skill → daily return. The gate warns, then yields.¹²



THE HYPOTHESIS NEEDED THE DASHED BOXES. THEY WERE NEVER SHIPPED.

High / low, defined up front.

Measure challenge lessons completed

Not days coming back *is* retention — circular

Cut at 3 where the D7 curve steps — 18%
→ 39%

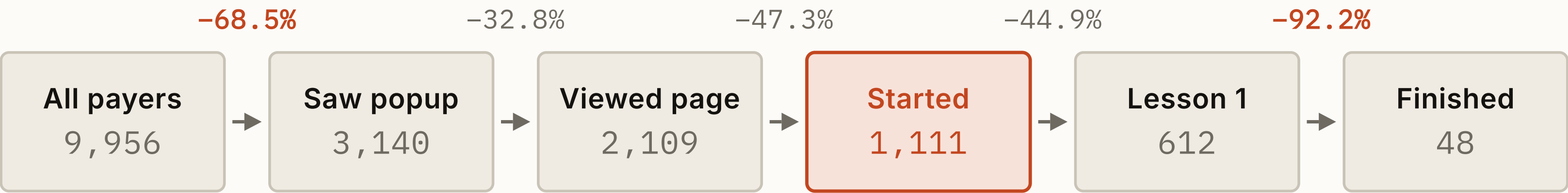
Robust swept every threshold 1–8 ¹³

TIER	RULE	USERS	SHARE
High	≥ 3 of 8 lessons	158	1.6%
Low	≤ 2 lessons	953	9.6%
Didn't take it	never started	8,845	88.8%

89% never start it.

But 94.6% of those who see the popup click it — distribution, not desire.¹⁴

LOSS AT EACH TRANSITION



TWO CLIFFS: NEVER OFFERED IT, AND PRESSED START THEN GOT NOTHING

Reads like the Challenge triples D1.

METRIC	HIGH · 158	LOW · 953	DIDN'T TAKE · 8,845
D1 — the target	69.6%	58.4%	21.7%
D3	57.0%	34.4%	10.0%
D7	39.1%	18.0%	6.9%
Unsubscribe	11.4%	11.4%	14.3%
Lessons, all courses	19.28	6.38	1.81
CSAT	3.33	3.53	3.74

It does not. Same clock — first app day. Intervals in the appendix.¹⁵

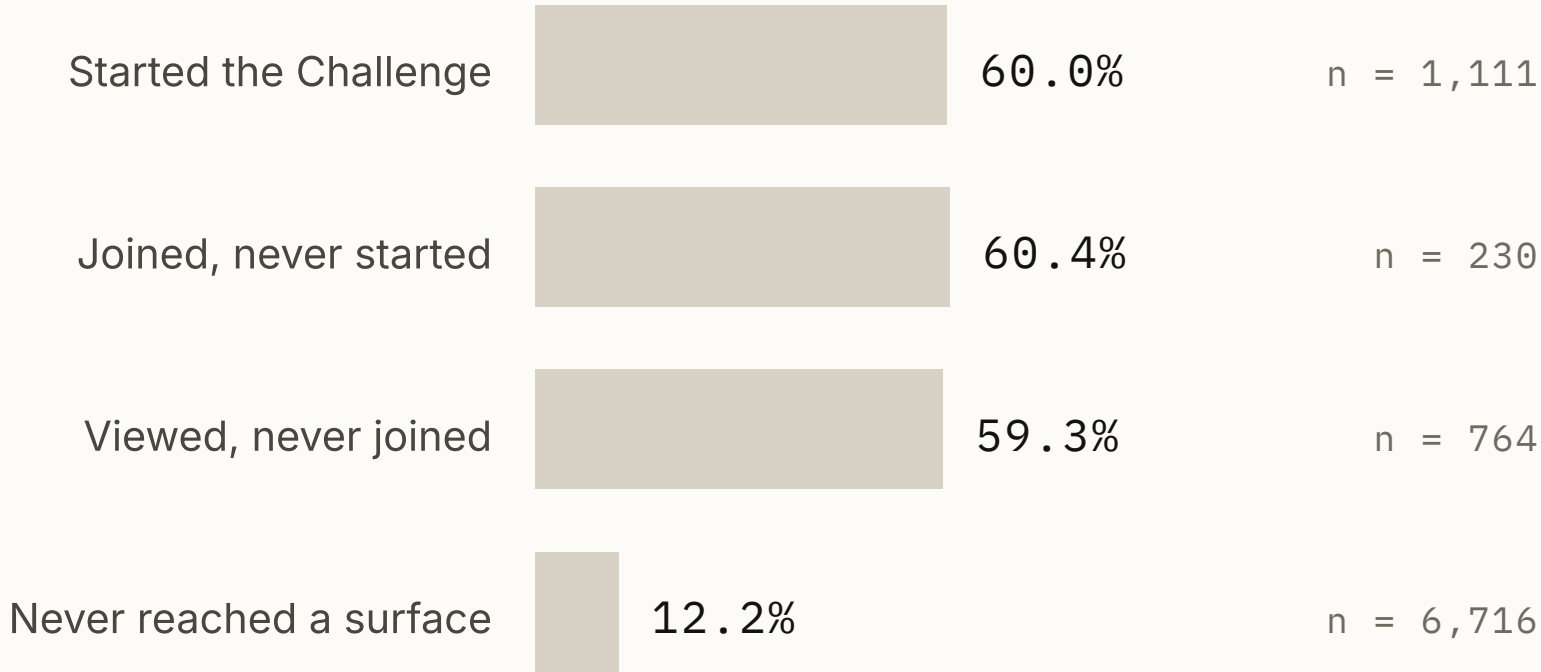
Same screen. Same result.

- The 8,845 mostly never saw the feature — not a control
- The fair test takers vs people who saw it and declined
- Read the bars top three are level · the cliff is the bottom one

+0.5

PERCENTAGE POINTS · P = 0.82¹⁶

D1 RETENTION · COMMON CLOCK



No segment rescues it.

Take-up range 8.8% – 14.9% across every cut

Taker gap roughly constant across ages¹⁷

AGE	BUYERS	TAKE RATE	D1 TAKERS	D1 NON-TAKERS
55+	32.1%	10.2%	60.9%	21.4%
45–54	25.6%	10.9%	62.1%	22.6%
35–44	19.0%	12.0%	66.1%	22.5%
25–34	11.2%	12.4%	54.0%	20.1%
18–24	9.2%	10.3%	44.7%	19.3%

WORK STATUS	BUYERS	TAKE RATE
Full-time employee	33.6%	11.6%
Business owner	18.3%	8.8%
Freelancer	14.9%	11.6%
Between jobs	5.8%	13.1%
Student	1.8%	14.9%

You measure something else.

Brief says D1. Your hierarchy says D1 is **secondary**.¹⁸

Primary gross profit · unsub % 12h/24h · rebill 0 → 1, 1 → 2 · LTV

Observable here unsub timing yes · rebill no, 17 events · profit no¹⁹

Secondary D1 / D3 / D7 · session · CSAT — what it was aimed at

So scoped second-order, and missed that too

No.
Not a success.

D1 — target +0.5 pts vs matched control · p = 0.82

Unsubscribe -2.3 pts · p = 0.12 · not significant

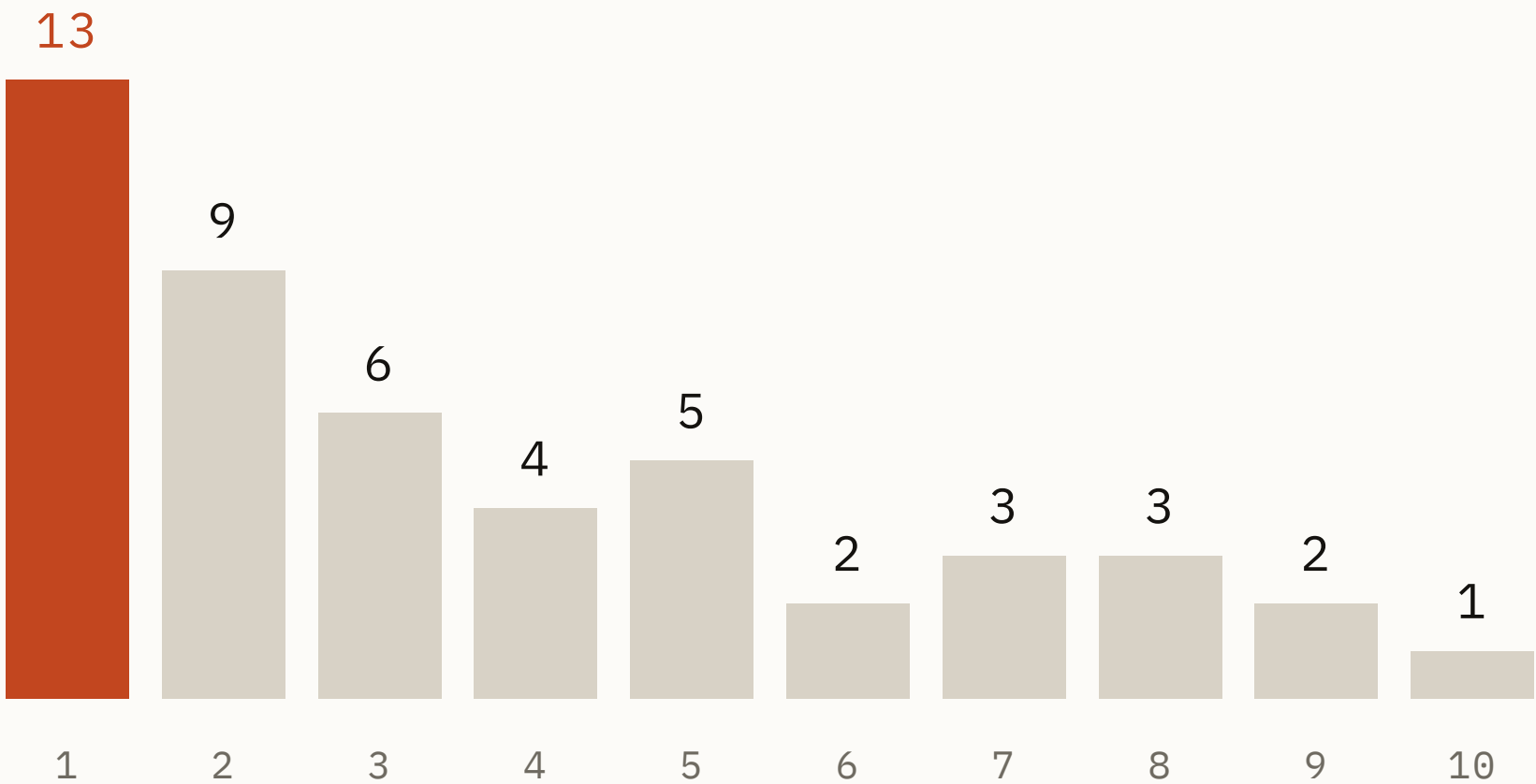
CSAT falls as engagement rises · 3.74 → 3.10 ²⁰

Rule set first dose-response surviving exposure
 matching — failed both

A soft gate is not a gate.

- 27% finished in a single sitting
- 46% within two days
- 1.46 mean lessons per active challenge day
- Verdict** not disproven — never tested

DAYS TAKEN TO FINISH THE “7-DAY” CHALLENGE · N = 48²¹



It isn't unappealing. It's hidden.

PHASE	SHIPS	METRIC IT MOVES
0 · Placement	Open on the Challenge · 80/20 hold-out	Biggest loss — and it gives you the control group
1 · First 10 min	Why 45% get nothing · guaranteed win	Unsub % 12h/24h — primary
2 · Mechanic	Make the gate real · wire the reminder	Rebill 0 → 1 — primary
3 · Hand-off	Day ends → send them to the plan	D3, D7 — today it ends nowhere
4 · Quality	Mine 3,601 exit-reason events	CSAT — guardrail 3.45
Why placement first 94.6% of people shown it clicked · 68.5% were never shown it		
Kill rule hold-out ships · no lift · kill it ²²		

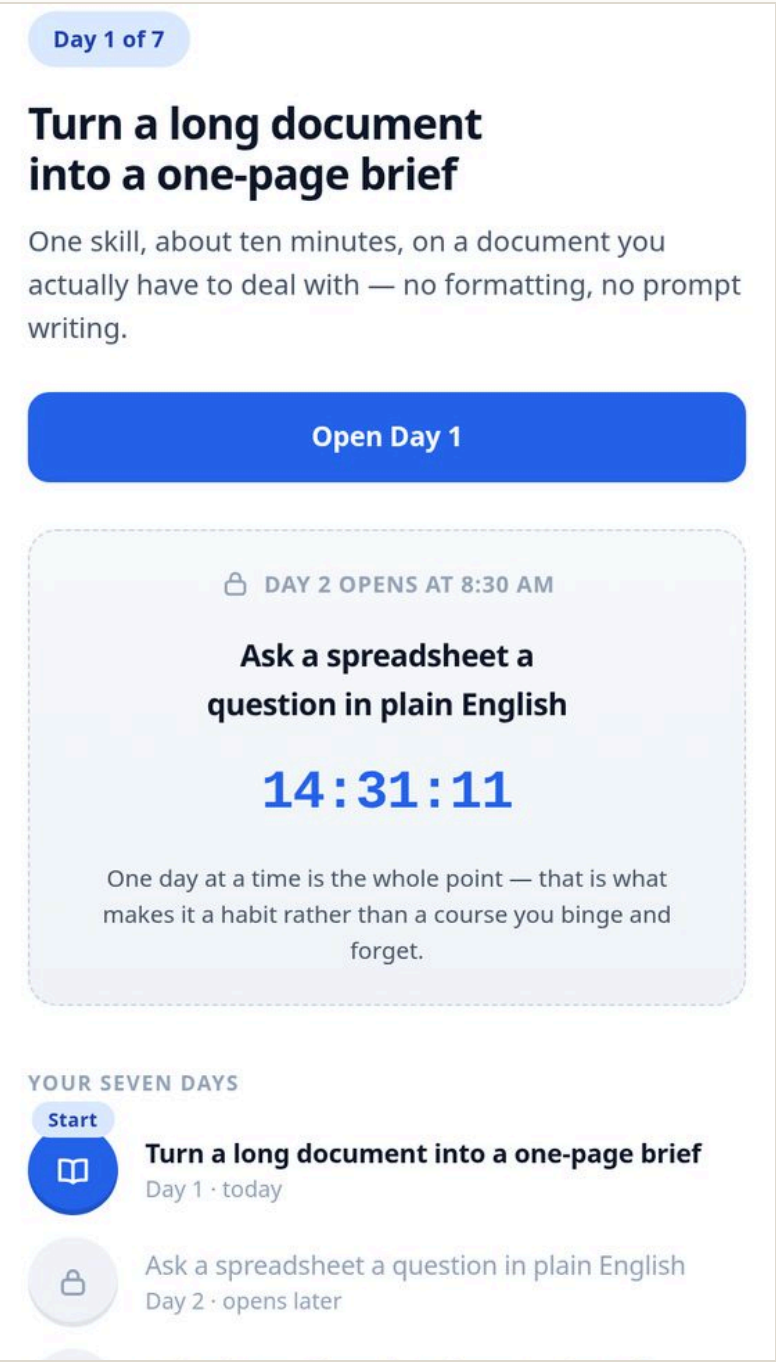
The loop, built.

The hook the Challenge is what greets you

The grind the plan — open, no gate

The missing arrow day ends → you land on the plan

alyasska.github.io/Nomad_Venture_Studio_TA_C4



1 · Enter
the Challenge, served on arrival

2 · Day 4
checks you cannot skip past

3 · Hand-off
tomorrow locked → go to your plan

4 · Plan
the track with no gate

PART D

Subscription economics

Blended net LTV
\$125.06.

PLAN	MIX	GROSS	NET
1-week	10%	\$68.66	\$60.42
4-week	70%	\$140.57	\$123.70
12-week	20%	\$184.26	\$162.15
Blended	100%	—	\$125.06

Survival S_k = running product of the C-curve

Gross intro + recurring $\times \Sigma S_k$ + both upsells

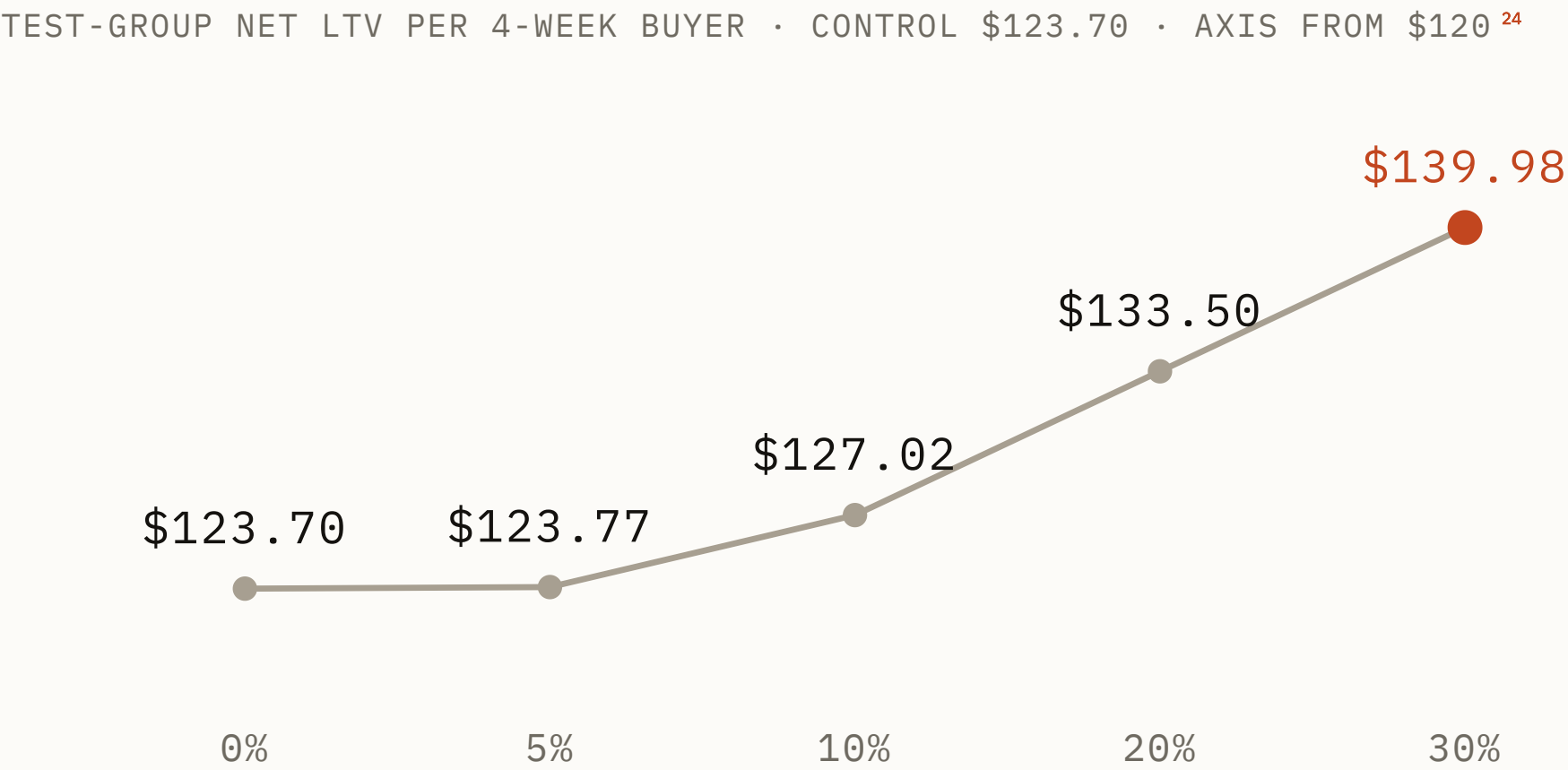
Net $\times 0.88$ — 12% fee on all collected cash

Validated observed mix 10.2 / 64.6 / 25.1 $\rightarrow$ \$126.90²³

Weak point 1-week plan · 34.6% churn in 14 days

Break-even 4.9%. Run it.

Each upgrade	+\$49.99 + \$23.69 recurring = \$73.68
Cannibalises	\$3.60 second upsell
p*	$3.60 \div 73.68 = 4.88\%$
Conservative	8.28% on the 52-week horizon
At risk	\$3.60 per test-arm buyer



LTV prices. It cannot detect.

Your words: LTV rarely reacts to anything but the obvious. Here is the size — so read a v2 on unsub 12h/24h + rebill, then price it with LTV. ²⁵

+\$1.31

PER +1 PT OF REBILL 0→1
≈1% of blended LTV — inside cohort noise

\$478k

SAME POINT, PER YEAR
at 1,000 signups a day

\$1.69m

+2 PTS ON BOTH REBILLS
money large · signal faint

Every task, one line.

A1	6 segments on emotion + JTBD · ~60% of buyers 45+ · churn falls with age
A2	Acute risk is the refund — 11.3% of unsubs are payment failures or disputes
B1	4 direct twins · the dangerous indirect competitor is free content
B2	They win on containers, practice surfaces, paywall craft — not content
B3	5 steals · de-risk → retain → differentiate → diversify → expand
C1	89% never start · 45% of starters do zero lessons · gap real, not causal
C2	Not a success · +0.5 pts, p = 0.82 · never tested its own hypothesis
C3	Rebuild the mechanic · instrument first · reach last
C4	Working prototype of the daily gate · pre-registered bar · kill rule
D1	Net LTV \$60 / \$124 / \$162 · blended \$125.06 · validated on real mix
D2	Break-even 4.9% · run it · downside bounded to \$3.60 per buyer

Three the data cannot answer.

01 **Was D1 ever the real bar?**

no hold-out exists · if it shipped as a refund gate, my verdict changes

02 **What D1 = success?**

I can say it moved nothing · not whether it missed by a little

03 **Which LTV horizon?**

full curve or 52 weeks · worth ~\$5 of blended LTV

Sources.

Re-run

python3 05_qa.py

Result

91 checks · any drift fails

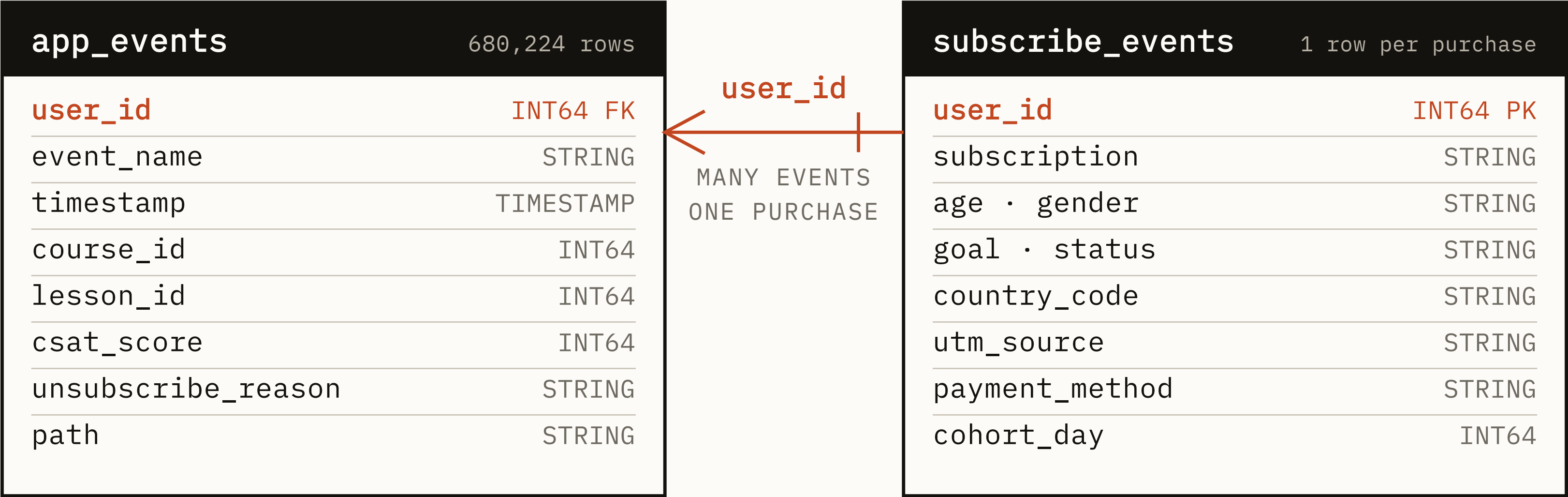
1	Reproducibility harness part-c-release-verdict/analysis/05_qa.py ↩ 03	11	Prioritized recommendations part-b-competitors/03-recommendations.md ↩ 14	23	LTV model + plan-mix validation part-d-economics/model/ltv_model.py · analysis/04_loose_ends.py ↩ 29
2	50.2% never complete a lesson analysis/04_loose_ends.py · 4,999 of 9,956 ↩ 04	12	Feature = course_id 338 · 99.7% of all challenge starts app_events · analysis/01_diagnostics.py Q5 ↩ 17	24	A/B break-even model part-d-economics/02-ab-test-model.md ↩ 30
3	Every app_events user has a purchase row sql/13_comparison_groups.sql ↩ 05	13	Tier definition + threshold sweep analysis/02_main.py §2, §4 ↩ 18	25	LTV sensitivity to rebill part-d-economics/model/ltv_sensitivity.py ↩ 31
4	k-modes clustering, 38,071 respondents part-a-audience/analysis/cluster_quiz.py ↩ 07 · 47	14	Exposure ladder, 9,956 payers analysis/02_main.py §1 · sql/13_comparison_groups.sql ↩ 19	26	Stratified comparison analysis/04_loose_ends.py ↩ 37
5	Buyer demographics, n = 9,956 subscribe_events · analysis/02_main.py §6 ↩ 08	15	Tier comparison, Wilson intervals analysis/02_main.py §3 ↩ 20	27	Immortal-time correction analysis/02_main.py §4 Test B ↩ 38
6	Self-serve cancel used 1,358 times app_events · subscription_manage_subscription_unsubscribe_click ↩ 09	16	Exposure-matched control · z = +0.22 analysis/02_main.py §4 Test A ↩ 21	28	Non-tautological dose-response analysis/02_main.py §4 Test C ↩ 39
7	11.3% of unsubs are payment / chargeback / dispute app_events.unsubscribe_reason · analysis/03_supplement.py §E ↩ 10	17	Segment behaviour analysis/02_main.py §6 ↩ 22	29	Target metric under both anchors analysis/03_supplement.py §A ↩ 40
8	FTC v. Genesis, June 2026 research/deep-05-product-reality.md ↩ 10	18	Metric hierarchy · Jobescape PM, 2026-07-28 00-context/pm-conversation.md ↩ 23	30	CSAT inversion analysis/03_supplement.py §B · 3,740 / 845 / 158 / 48 raters ↩ 41
9	Competitor set and classification part-b-competitors/01-competitors.md ↩ 12	19	subscription_renewed = 17 events, n = 9,956 app_events.event catalogue · sql/14_unsub_timing.sql ↩ 23	31	Automation tab and streak, my paid account part-a-audience/materials/walkthrough/observations.md ↩ 43
10	Competitor teardowns part-b-competitors/02-analysis.md · research/deep-01, deep-02 ↩ 13 · 15	20	CSAT by engagement tier analysis/03_supplement.py §B ↩ 24	32	Full tier table deliverable/appendix/02_main-output.txt ↩ 45
		21	Days-to-finish distribution, 48 finishers analysis/02_main.py §5 ↩ 25	33	Data-quality flags analysis/03_supplement.py §E ↩ 46
		22	v2 plan, phases and kill rule part-c-release-verdict/03-whats-next.md ↩ 26		

APPENDIX

The working behind the answers

Two tables. One key.

The product sits entirely behind the paywall.



EVERY app_events USER HAS A PURCHASE ROW – "USER" AND "PAYER" ARE THE SAME SET

Activity held constant → effect stays zero.

PRODUCT ACTIVITY	N TAKERS	D1 TAKERS	N CONTROL	D1 CONTROL	DIFF	P
1 active day	97	0.0%	63	0.0%	+0.0	1.000
2	270	46.3%	257	42.4%	+3.9	0.370
3-4	446	64.8%	427	65.1%	-0.3	0.924
5-7	217	82.9%	205	84.4%	-1.4	0.689
8+	81	90.1%	42	76.2%	+13.9	0.038

4 of 5 strata flat · the significant cell is smallest and 1 test in 5.²⁶

Takers are pre-selected survivors.

24% start on day 0

Median day +1

28% day +3 or later²⁷

+38.3

RAW GAP
all takers vs never-took

+17.4

– SURVIVAL HEAD START
day-0 starters only

+0.5

– EXPOSURE SELECTION
vs looked-and-left

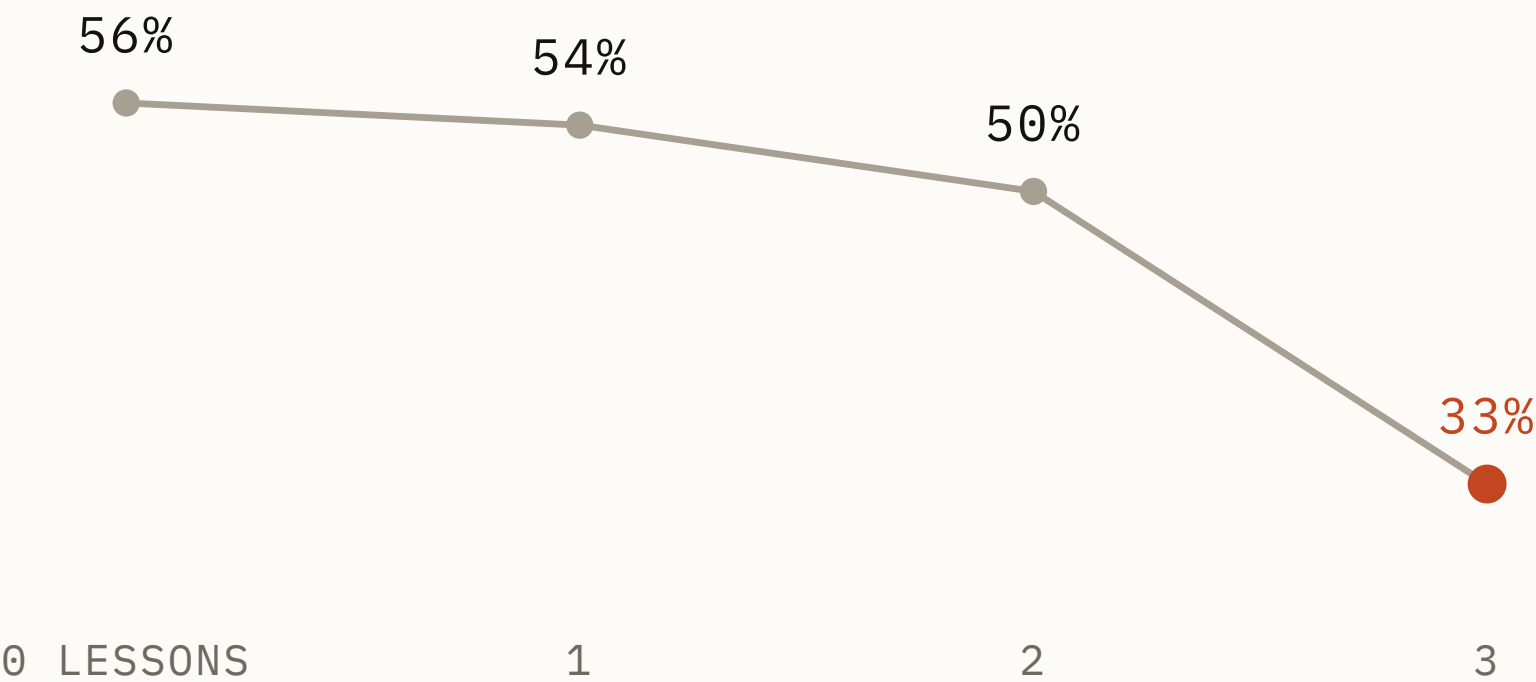
Bigger day-0
dose → *worse*
return.

Clean test 816 takers · all activity on one day

Result -9.2 pts · p = 0.048

Bingers all 8 in a day → 38.5% unsub

D1 NEXT DAY, BY DAY-0 DOSE · AXIS FROM 28% ²⁸



Your dashboard would call this a win.

39.8% vs a 26.0% baseline = a **+14-point win** — an artifact. ²⁹

Different clocks takers from challenge-start · others from first app day

Different populations 88.8% of the book · ⅔ never reached a surface

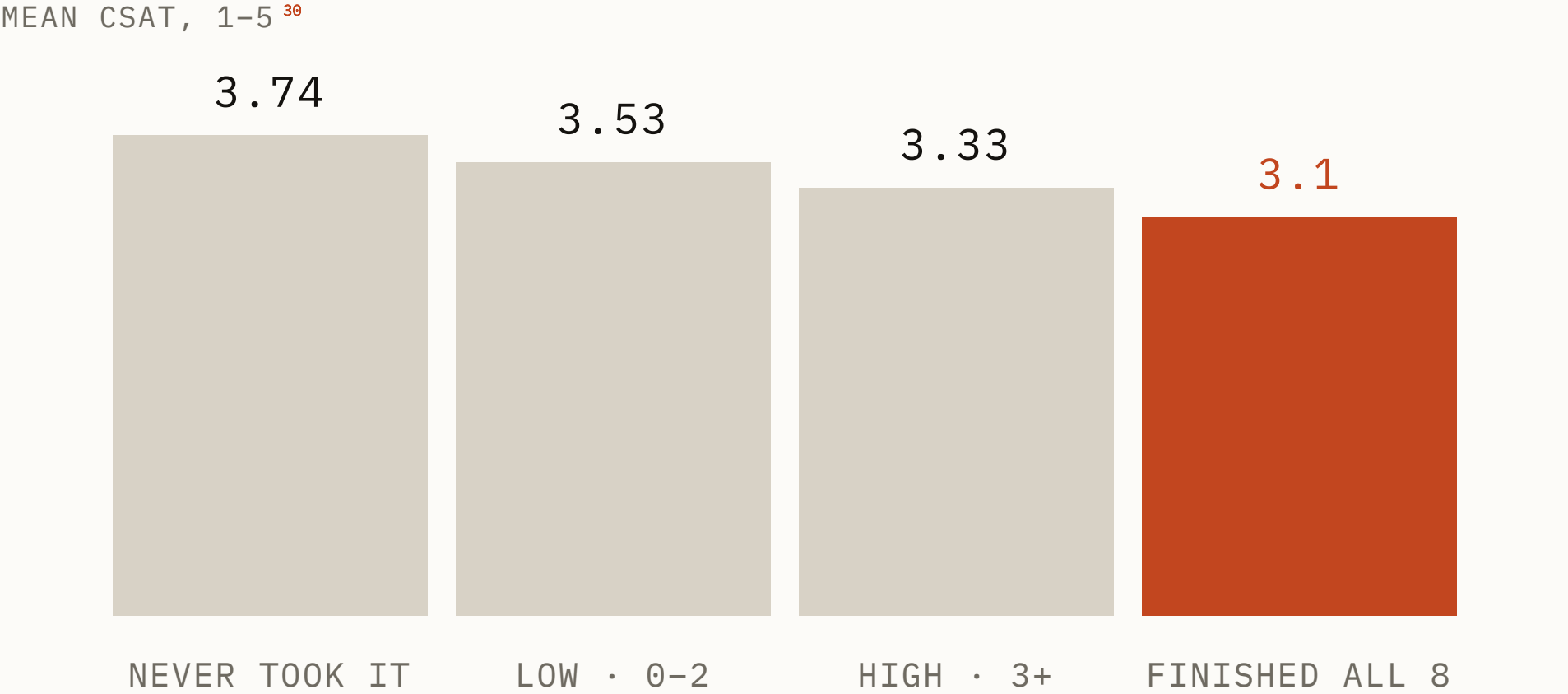
No hold-out shipped to 100%

Corrected +38 → +17 → +0.5

Satisfaction falls as engagement rises.

Challenge content	3.25
Same users, elsewhere	3.45

So scaling reach on 3.25-star content scales refunds



Pre-registered success criteria.

	METRIC	BAR
Primary	Unsub % 12h / 24h, test vs hold-out	the business metric
Primary	Rebill 0 → 1	~\$1.31 blended LTV per point
Secondary	Exposure-matched D1 lift	≥ +5 points
Secondary	Starters completing ≥1 lesson	55% → 75%
Guardrail	Challenge CSAT	≥ 3.45 · must not fall
Guardrail	Refund / chargeback rate	must not rise
Guardrail	Lessons per starter	must not fall — gating risk
Stated before the test, so the result cannot be reinterpreted after.		

The rebuild is a wiring job.

Already ships

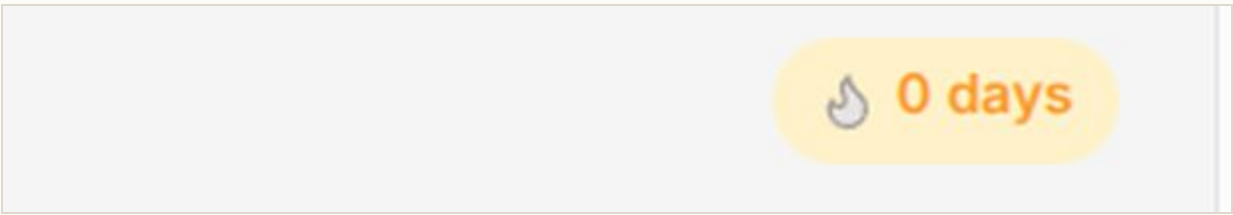
WhatsApp agent · motivational messages · study reminders

Already renders

streak counter · weekly tracker

Shows

0³¹



THE STREAK, IN PRODUCTION

What I used, and why.

No regression, no model — a **natural control**.

- 01 **Wilson intervals**
small cells · proportions near 0 and 1
- 02 **Two-proportion z-tests**
pooled variance · diff, z, p reported
- 03 **Explicit denominators**
D7 observable for 72.5% only
- 04 **Multiple comparisons**
1 significant stratum in 5 · reported as such

Tiers, intervals, denominators.

METRIC	HIGH	LOW	DIDN'T TAKE	N · H / L / D
D1 — target	69.6% [62.1–76.3]	58.4% [55.3–61.5]	21.7% [20.9–22.6]	158 / 953 / 8,845
D3	57.0% [49.2–64.4]	34.4% [31.5–37.5]	10.0% [9.4–10.6]	158 / 953 / 8,845
D7	39.1% [31.2–47.6]	18.0% [15.4–21.0]	6.9% [6.3–7.5]	133 / 737 / 6,345
Unsubscribe	11.4% [7.3–17.3]	11.4% [9.6–13.6]	14.3% [13.6–15.1]	158 / 953 / 8,845
Lessons, all courses	19.28	6.38	1.81	—
Active days	5.37	3.46	1.76	—
CSAT · raters	3.33 (158)	3.53 (845)	3.74 (3,740)	—
95% Wilson · anchored on each user's first app day. ³²				

Flags before anyone acts.

Two are severe enough to state before any decision. ³³

01	No hold-out group — starts on every day of the window · no control arm
02	14-day window — D7 unobservable for 27.5% · unsub rates are floors
03	Two quiz vocabularies pooled — “Full-time worker” 28.5% vs “employee” 11.8%
04	Popup carries no challenge id — 338 is 99.7% of starts — impact negligible
05	“7-Day Challenge” ships 8 lessons — promise and payload off by one

I asserted six. Then I tested it.

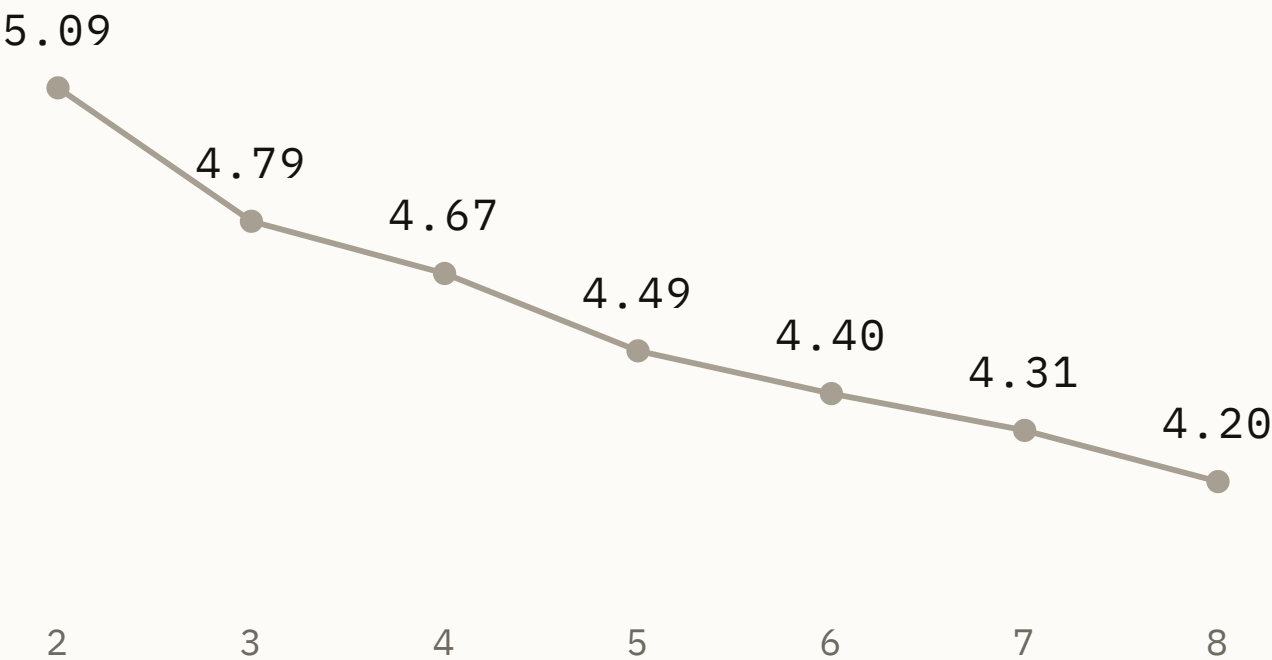
Method k-modes · Hamming · per-attribute modes.
Not k-means — a mean over “35–44” has no meaning

Sample 38,071 respondents × 10 categorical answers

Finding no elbow — cost falls smoothly, so the population is a continuum, not natural kinds⁴

Six is a resolution I chose and defended — not a count the data forced. Two authored segments (Gen Z, Burned Doer) did not survive; one real cluster (The Adept, 20.9%) was missing from the framework entirely.

WITHIN-CLUSTER COST PER USER, K = 2 ... 8



One row per user.

Both anchors in one query — that is what makes the matched comparison possible.

```
ret AS ( -- retention off BOTH anchors, so the groups stay comparable
  SELECT a.user_id,
         MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 1 DAY),1,0)) AS ret_d1,
         MAX(IF(dd.d = DATE_ADD(a.first_seen,  INTERVAL 1 DAY),1,0)) AS fs_ret_d1

  CASE WHEN a.ch_start_date IS NOT NULL THEN 'took_338'
        WHEN a.ch_joined = 1                THEN 'joined_338_never_started'
        WHEN a.ch_viewed  = 1                THEN 'viewed_338_never_joined'
        ELSE                                'no_challenge' END AS group_338
```